

PREDICTING TOTAL CALORIES BURNED

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ABSTRACT

This project investigates the use of supervised machine learning techniques to predict the total calories burned during exercise using a lifestyle dataset containing 20,000 observations and over 50 features. Machine learning provides a powerful framework for modeling the complex nonlinear relationships that influence human calorie burn. The objective of this study was to develop, evaluate and compare four models using a preprocessing and feature-engineering pipeline designed to prevent data leakage and enhance model interpretability. The models used include Support Vector Regression (SVR), Random Forest (RF), AdaBoost, and Gradient Boosting.

A comprehensive exploratory analysis of the data revealed that behavioural workout-related variables were very predictive of calorie expenditure in comparison to physiological measures such as heart rate. After transforming categorical variables, normalizing continuous features, and preparing the final predictors, each model was trained and testing using an 80/20 split. SVR demonstrated stable linear predictive behavior, but its accuracy was limited by the use of integer-encoded categorical features and the model's overall capacity. Random Forest provided robust performance and captured complex interactions, though prediction variance increased slightly for very high calorie sessions. AdaBoost produced reasonable accuracy but consistently underpredicted at higher calorie rates due to shallow weak learners. Gradient Boosting was also an effective model by achieving low error rates and good calibration between predicted and actual values.

The results confirm that machine learning can provide highly accurate and reliable calorie-burn predictions from rich, multi-feature datasets. These findings highlight the potential for integrating machine learning models into personalized fitness applications, workout tracking systems and digital health platforms.

INTRODUCTION

Understanding human energy expenditure has become an increasingly important topic within health sciences, sports analytics, and personalized fitness applications. Calorie burn prediction supports a wide range of decisions such as optimizing personal workout programs and improving clinical assessments of metabolic health. Accurate estimation of calories burned can assist individuals in tailoring nutrition plans, maintaining energy balance, and achieving weight management goals. Traditionally, caloric expenditure has been measured using metabolic carts, wearable sensors, or heart-rate monitors linked with empirical formulas. However, these approaches can be expensive, inaccessible, or limited in accuracy across different populations and activity types.

The growth of machine learning (ML) provides new opportunities to model complex, nonlinear relationships between physiological traits, lifestyle habits, exercise parameters, and energy

consumption. Recent studies demonstrate that ML-based predictors often outperform classical regression formulas (e.g., Harris-Benedict or MET-based estimations) because they can capture interactions among variables such as age, body composition, workout intensity, macronutrient intake, and exercise modality (Srihari et al., 2022). Additionally, ML models can adapt more effectively to high-dimensional datasets generated by modern digital health platforms and fitness trackers.

In this project, a lifestyle and exercise dataset sourced from Kaggle, “[Life-Style Data](#)” by Oikarinen (2023), serves as the basis for the development of a calories-burned prediction model. The dataset contains 20,000 entries and over 50 features describing participant demographics (e.g., age, gender, weight, height), biometrics, workout characteristics (such as workout type, duration, intensity), and nutrition-related variables. This combination of features presents an ideal case for machine learning-based regression since calories burned is influenced by multiple correlated and nonlinear physiological processes.

The scientific foundation for energy-expenditure modeling is well established. Previous studies demonstrate that heart-rate characteristics (such as average BPM and heart-rate reserve) are strong predictors of caloric output because they reflect internal workload and cardiovascular response (Keytel et al., 2005). Similarly, workout duration and intensity directly influence total energy expenditure. Body composition measures—including weight, fat percentage, and lean muscle mass—also play a significant role because they determine the basal metabolic rate and influence the energy cost of physical activity (Thompson et al., 2010). More recent research highlights the potential for machine learning and multi-sensor data to outperform traditional metabolic equations. For example, Shcherbina et al. (2017) compared consumer wearables and demonstrated that even commercial devices showed large variation in energy-expenditure estimation, suggesting that personalized, data-driven models may offer improved accuracy. With the growth of lifestyle tracking and health technology, machine learning has become increasingly important for interpreting complex physiological and behavioral datasets.

The primary objective of this study is to develop, evaluate, and compare four supervised regression algorithms: Support Vector Regression (SVR), Random Forest (RF), AdaBoost, and Gradient Boosting. Rather than relying on a single approach, this comparative analysis seeks to characterize how different learning strategies, from kernels to ensemble techniques, perform on high-dimensional lifestyle data.

Beyond predictive performance, this study also emphasizes the critical role of feature engineering and data preprocessing in machine learning pipelines. By rigorously cleaning data and preventing leakage, the project aims to identify which behavioral and physiological features most strongly drive calories expenditure. Ultimately, these findings highlight the potential for integrating sophisticated ML models into digital health platforms, offering individuals more accurate, personalized insights for weight management and training optimization.

DATA

Dataset

The following lifestyle dataset was used for this project:

<https://www.kaggle.com/datasets/jockeroika/life-style-data/data>

This study uses a comprehensive lifestyle and fitness dataset containing 20,000 observations and 54 variables related to demographic characteristics, physiological measurements, workout habits, nutrition, and daily lifestyle patterns. The primary objective is to develop machine learning models capable of predicting Calories Burned during a workout session.

The dataset includes the following groups of variables:

Table 1. Dataset variables and their categories

Category	Features Included
1. Demographic C Anthropometric	Age, Gender, Height (m), Weight (kg), BMI, BMI_calc, Fat_Percentage, Lean_mass_kg
2. Heart Rate C Physiological Metrics	Max_BPM, Avg_BPM, Resting_BPM, pct_HRR, pct_maxHR
3. Workout C Exercise Characteristics	Session_Duration (hours), Workout_Type, Workout_Frequency (days/week), Experience_Level, Name of Exercise, Sets, Reps, Benefit, Workout, Difficulty Level, Body Part, Type of Muscle, Target Muscle Group, Equipment Needed
4. Calorie Burn C Exercise Energy Metrics	Calories_Burned (target variable), Burns Calories (per 30 min), Burns Calories (per 30 min)_bc, expected_burn
5. Nutrition C Dietary Intake	Daily meals frequency, meal_name, meal_type, diet_type, Carbs, Proteins, Fats, Calories, cal_from_macros, pct_carbs, protein_per_kg, sugar_g, sodium_mg, cholesterol_mg, serving_size_g, cooking_method, prep_time_min, cook_time_min, rating
6. Hydration C Lifestyle Habits	Water_Intake (liters), Physical exercise, cal_balance
7. Other / Combined Indicators	Name of Exercise, Workout, Benefit (if needed), Burns_Calories_Bin (categorical target version)

`Calories_Burned` is the primary output variable that the machine learning models aim to predict. It represents the estimated number of calories burned during a workout session based on physiological and activity-related features.

Exploratory Data Analysis (EDA)

Distribution of the Target Variable

A histogram and boxplot of `Calories_Burned` show:

- Most workout types burn between 900 and 1600 kcal.
- The distribution is right-skewed, with high-intensity workouts reaching 2,500-2,900 kcal.
- Outliers exist at the high end but appear valid given the context.

This confirms a wide variance in energy expenditure, likely driven by session duration and workout intensity.

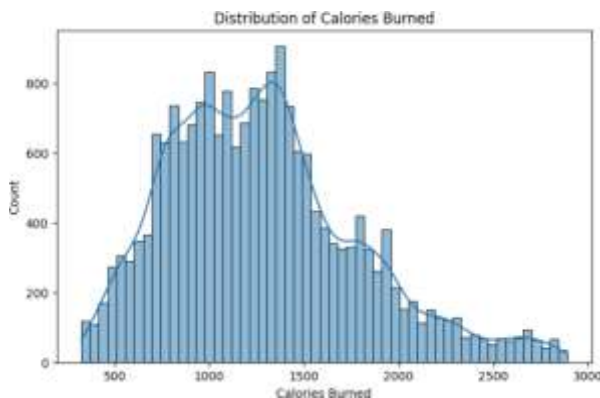


Figure 1. Distribution of Calories Burned

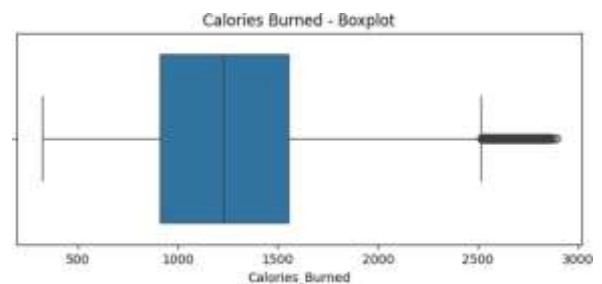


Figure 2. Calories Burned Boxplot

Distribution of Numeric Features

Histograms for 11 core numerical variables (Age, Weight, Height, heart rate features, session duration, workout frequency, fat percentage, lean mass) show:

- Age: Uniformly distributed between 18 and 60.
- Height & Weight: Normal-like distributions centered around typical adult fitness ranges.
- Heart Rate Variables: Healthy exercise zone patterns.
- Workout Frequency: Centered around 3-4 days/week.
- Session Duration: Most workouts last 1-1.5 hours.
- Body Composition Variables: Realistic spreads for Fat Percentage (15-35%) and Lean Mass.

No unrealistic values were detected.

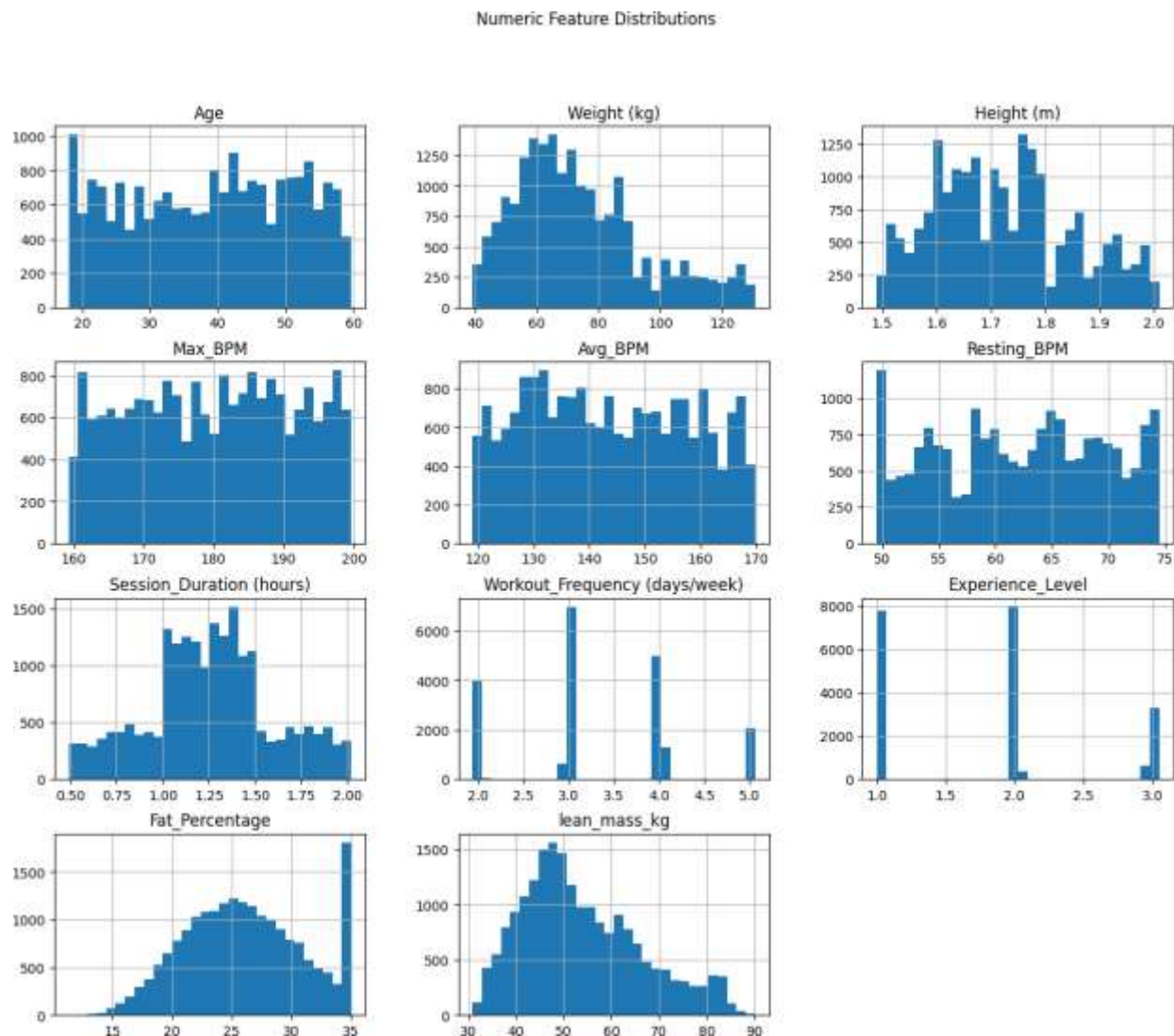


Figure 3. Numeric Feature Distributions

Correlation Analysis

A correlation heatmap was generated for all numeric variables.

Table 2. Top correlations with Calories Burned

Feature	Correlation
Session_Duration (hours)	0.81
expected_burn	0.77
Experience_Level	0.70

Workout_Frequency	0.58
Water_Intake	0.26

According to the correlation values, Session Duration is the strongest driver of caloric expenditure. More experienced and frequent exercisers burn more calories. Heart rate features (Avg_BPM, Max_BPM) showed surprisingly weak correlations, suggesting they require interaction with other variables (e.g., session duration) to become meaningful.

Scatterplots

Scatterplots between Calories Burned and Avg_BPM, Max_BPM, Resting_BPM revealed no strong visual trends, confirming low correlations. However, Session Duration vs Calories Burned shows a clear positive linear pattern. This supports the hypothesis that behavioral features outperform physiological features for caloric prediction in this dataset.

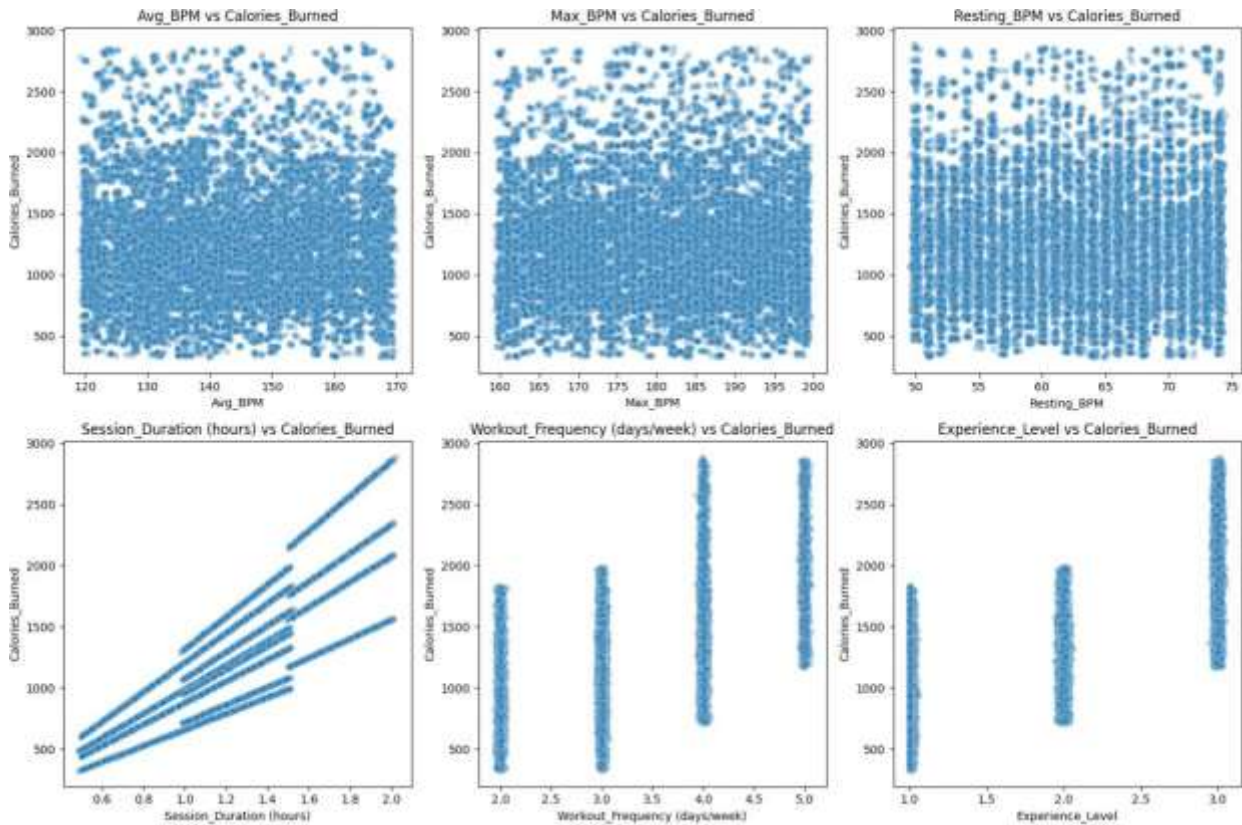


Figure 4. Scatterplots between calories burned

Categorical Variable Exploration

Bar plots of average Calories Burned across categories revealed that Workout Type strongly influences energy expenditure (HIIT burns the highest average calories (~1650 kcal)), while Yoga

burns the least (~900 kcal). Gender has almost no effect (male and female values are nearly identical). Body Part trained shows minimal variation, suggesting session intensity matters more than muscle group targeted. Difficulty Level shows small differences, likely overshadowed by Experience Level. These findings highlight that workout characteristics play a more decisive role than demographic factors.

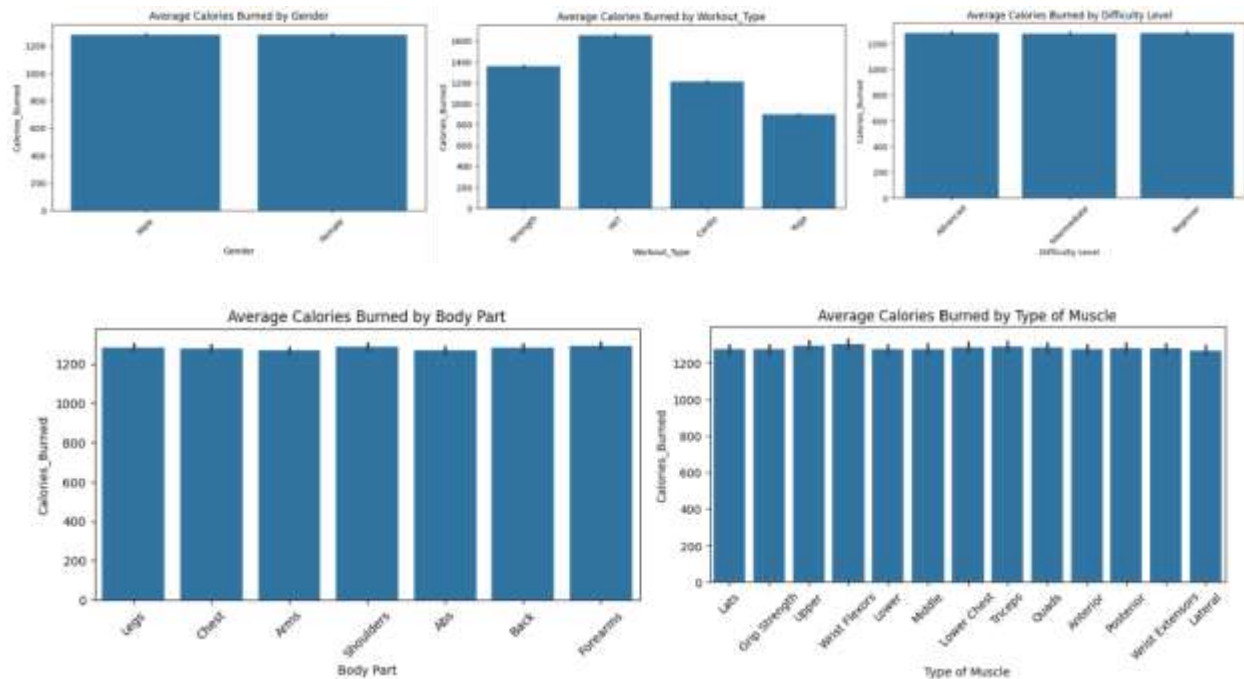


Figure 5. Bar plots of average calories burned

Outlier Detection Using Boxplots

Individual boxplots for all numeric variables revealed expected patterns:

- Height, Age, and heart rate metrics show very few outliers.
- Calories, sugar, sodium, cholesterol, and serving size display numerous outliers, which is typical for nutritional data.
- The combined standardized boxplot clearly identifies fields with extreme right-tail values (e.g., sodium, cholesterol, calorie density), but these values are realistic and not erroneous.

Since the goal is prediction rather than anomaly detection, outliers were kept to preserve real-world variation.

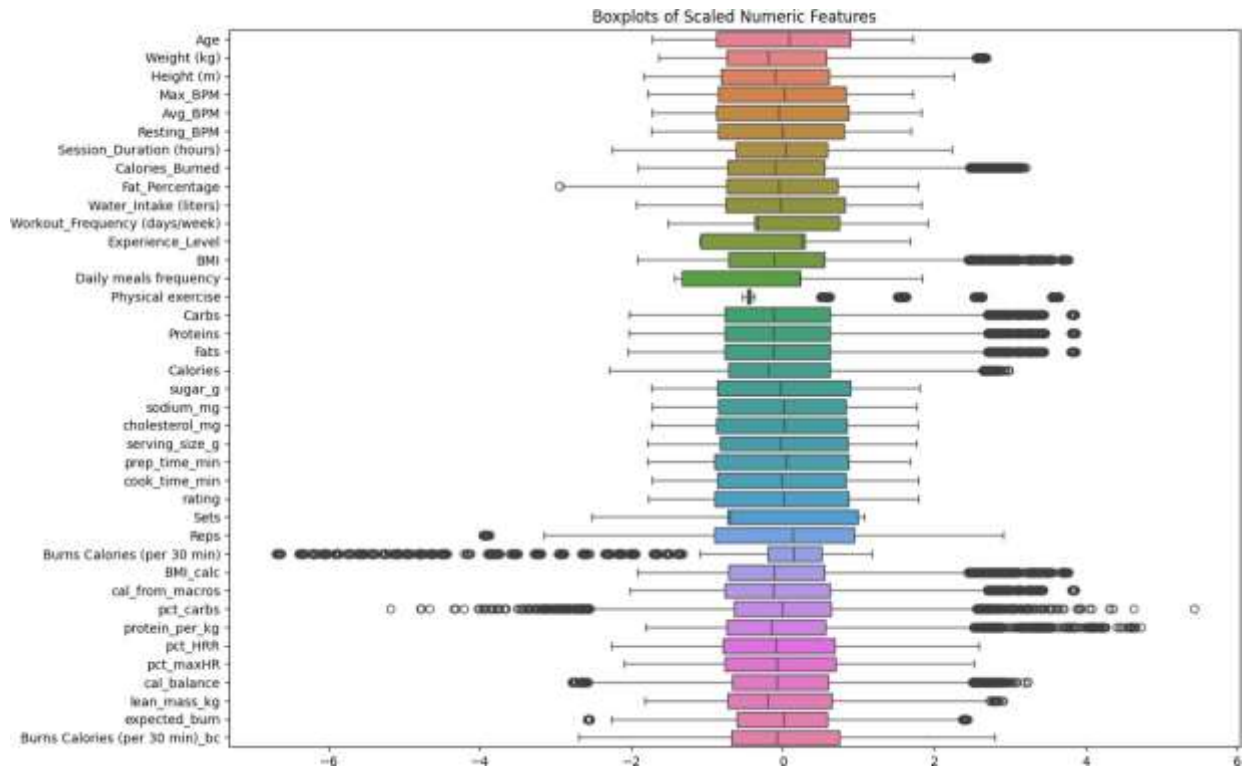


Figure 6. Boxplots of scaled numeric features

Feature Engineering, Data Cleaning and Leakage Prevention

The feature engineering process produced 46 usable predictors.

The following variables were removed because they directly reveal the target:

- expected_burn
- Burns Calories (per 30 min)
- Burns Calories (per 30 min) _bc
- cal_balance
- cal_from_macros

Removing them ensures the model must learn from true predictive features.

Columns such as meal_name, cooking_method, and specific exercise names were removed due to excessive category counts and low predictive consistency. Remaining categorical variables (Gender, Workout Type, Difficulty Level, Burns_Calories_Bin) were integer-encoded using LabelEncoder. The final modeling dataset includes:

- 46 engineered and cleaned features
- No leakage

- No excessively sparse categorical columns
- Normalized numerical values using StandardScaler
- Train-test split: 80/20

This structured dataset is well-prepared for model training using Gradient Boosting, XGBoost, LightGBM, Random Forest, and other algorithms.

METHODOLOGY

Support Vector Machine

The Support Vector Machine (SVM) methodology used in this project followed a structured machine learning pipeline designed to prepare the dataset, configure the model, and train a Support Vector Regression (SVR) algorithm for predictive analysis. SVM was selected for its strong theoretical foundation in handling nonlinear relationships and its ability to generalize effectively by maximizing the separation margin in high-dimensional feature spaces.

The processed dataset was divided into training and testing subsets using an 80/20 split to ensure a reliable estimate of generalization performance. The SVM model was implemented using the Support Vector Regression (SVR) class from the scikit-learn library. The RBF (Radial Basis Function) kernel was chosen, as it allows the model to project input features into a higher-dimensional space where complex nonlinear patterns can be more effectively modeled. Key hyperparameters including the regularization parameter C , which balances margin maximization and error tolerance; γ , which determines the influence of individual data points in the kernel space; and ϵ , which defines the margin of error within which no penalty is applied. These parameters were adjusted iteratively during experimentation to improve model behavior and reduce prediction variance.

The SVM training stage involved fitting the SVR model to the scaled training data and optimizing the hyperplane (or regression function) that best approximates the target variable while maintaining a low structural risk. After the model was trained, predictions were generated on the testing subset. This allowed the evaluation of the SVM's ability to generalize beyond the training data. Visual diagnostics, including predicted vs. actual plots, were also used to assess alignment between model outputs and ground truth values.

Random Forest

Random Forest (RF) is an ensemble learning method that constructs many decision trees during training and outputs the average predicted value across all trees. Each tree is trained on a different bootstrap sample of the dataset, while individual splits within each tree consider only a random subset of features. This combination of bagging and feature randomness reduces variance and

prevents the model from overfitting. Compared to individual decision trees, which tend to overfit over epochs, the ensemble approach produces a more stable and generalizable model. This is an important advantage for our dataset which contains a mix of categorical, continuous, and features with nonlinear relationships.

The implementation was performed using the scikit-learn machine learning library. Hyperparameters were optimized using *GridSearchCV*, which performs exhaustive search over specified configurations. *GridSearch* systematically evaluated combination of:

- *n_estimators*: [20, 60, 100, 200]
- *max_depth*: [None, 5, 10, 20]
- *min_samples_leaf*: [1, 2, 4]

GridSearchCV identified the configuration that yielded the lowest validation MAE (Mean Absolute Error). This final RF model was retrained automatically using the best parameters on the full training split and subsequently used to generate predictions for the test set.

Boosting

AdaBoost

AdaBoost (Adaptive Boosting) is an ensemble method that builds a strong model by combining many weak learners, typically shallow decision trees or stumps. The algorithm begins by assigning equal weights to all training samples. After each boosting step, it increases the weights of samples that were predicted poorly, forcing the next tree to focus more on the difficult observations. Each learner contributes to the final prediction with a weight proportional to its accuracy. Key parameters include the number of estimators and the learning rate, which controls how much each new tree affects the final model. The method reduces bias and improves predictive accuracy through sequential error correction.

The implementation is performed using the scikit-learn machine learning library. Hyperparameter tuning is conducted using *GridSearchCV*. The grid includes:

- *n_estimators*: [50, 100, 150]
- *learning_rate*: [0.01, 0.1, 1]

GridSearchCV searches all combinations within this hyperparameter grid using MAE as the scoring metric. The search identifies the configuration that yields the lowest validation error, and the best-performing AdaBoost model is automatically refitted on the full training split.

After tuning, predictions are generated on the test set using the best estimator returned by *GridSearch*. Performance is evaluated using MAE and Root Mean Squared Error (RMSE), which quantify the magnitude of prediction errors.

Gradient Boosting

Gradient Boosting is an ensemble learning method that builds a strong regressor by adding many weak learners in a sequential manner. Each new tree is trained to correct the residual errors of the combined model so far, effectively performing gradient descent in function space on a chosen loss function. This stage-wise optimization allows the model to capture complex nonlinear relationships while controlling overfitting through shallow trees and a small learning rate. Key hyperparameters include the number of estimators, learning rate, and the number of features considered at each split.

The implementation is performed using the scikit-learn machine learning library. A `GradientBoostingRegressor` is first initialized. Hyperparameter tuning is carried out using `GridSearchCV`, which systematically searches over a predefined grid. The hyperparameter grid is defined as:

- `n_estimators`: [50, 100, 150]
- `learning_rate`: [0.01, 0.1, 1.0]
- `max_features`: [0.5, 0.7, 1.0]

`GridSearchCV` evaluates all combinations in the grid and selects the configuration that minimizes the validation error. The best-performing Gradient Boosting model is then automatically refitted on the full training split.

Model performance on unseen data is assessed by generating predictions on the test set and evaluating them using MAE and RMSE. In a separate experiment using a manually specified model (`GradientBoostingRegressor (n_estimators = trees, learning_rate = 0.1)`), the Gradient Boosting regressor achieves an MAE of 17.98 and an RMSE of 23.99, indicating strong predictive performance on the target variable.

EXPERIMENTS C RESULTS

Support Vector Machine

After completing all the preprocessing steps, the Support Vector Regression (SVR) model was trained using the linear kernel, which was recommended by `GridSearch`. The experimental setup involved training the model on the standardized training set, followed by generating predictions on the test set to assess generalization performance.

To evaluate the SVR model, standard regression metrics were used, including MAE, RMSE, and the coefficient of determination (R^2). The R^2 score of approximately 0.71 indicates that the model explains a moderate portion of the variance in the target variable, while the MAE and RMSE values (around 230 and 276, respectively) reflect a noticeable level of prediction error.

The scatter plot of predicted versus actual values supports these observations. Although the points generally follow the diagonal trend, indicating that the model captures meaningful relationships, there is still considerable spread around the line. This demonstrates that the model's predictive accuracy is limited and that some patterns in the data remain unmodeled.

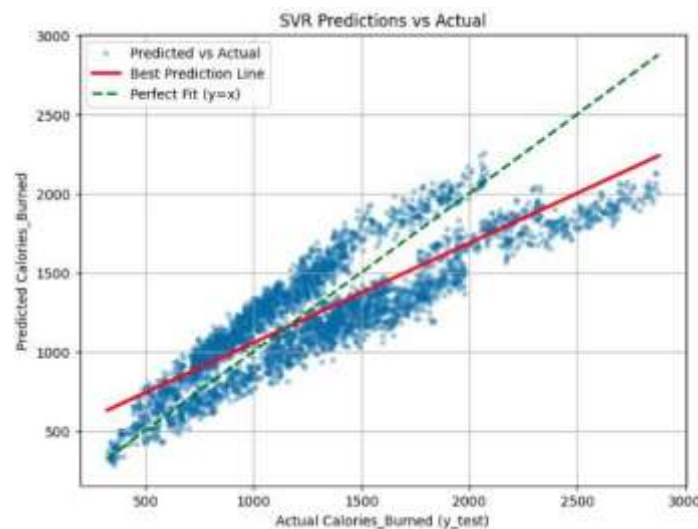


Figure 7. Actual Calories Burned vs Predicted Calories Burned Plot

The selected SVR configuration suggests that the underlying relationship between features and the target may be largely linear for this dataset. However, the performance also highlights the sensitivity of SVR to feature encoding and scaling, particularly given the presence of integer-encoded categorical variables. These encodings introduce artificial ordering and may restrict representational power.

Overall, the SVR model delivered reasonable but not highly accurate performance on the processed dataset. The results indicate that while linear-kernel SVR can extract useful patterns, further improvements could likely be achieved through alternative encoding strategies, expanded hyperparameter tuning, or exploring more flexible models such as tree-based regressors or boosted ensembles.

Random Forest

To assess the predictive capability of the RF model, we structured experiments around three key components: hyperparameter tuning, performance evaluation, and visualization of the fitted model's performance.

1. Hyperparameter Tuning

GridSearchCV evaluated $4 \times 4 \times 3 = 48$ combinations, each via 3-fold cross-validation. The search produced the following best configuration:

- `n_estimators = 200`
- `max_depth = 20`
- `min_samples_leaf = 1`

This configuration represents a moderately deep ensemble with a large number of trees, allowing the model to capture nonlinear patterns while ensure good generalization.

2. Cross-Validation Performance

The best average cross-validated error achieved was approximately 1.0005, which indicates the expected average error on unseen data during training.

3. Test Set Evaluation

Performance on the held-out test set was assessed Mean Absolute (MAE), Root Mean Squared Error (RMSE) and Predicted vs Actual scatter plot. The results are shown in the figures below.

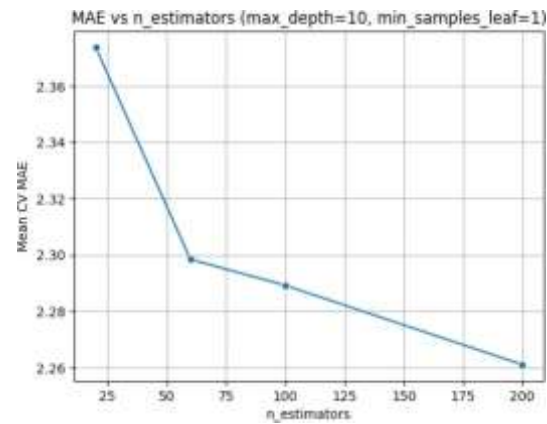


Figure 8. MAE vs. `n_estimators`

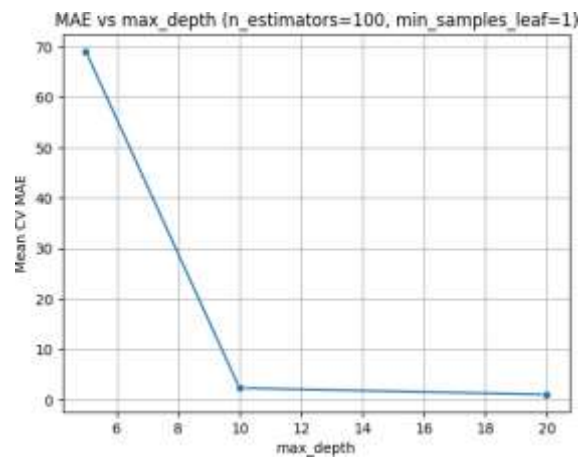


Figure 9. MAE vs. `max_depth`

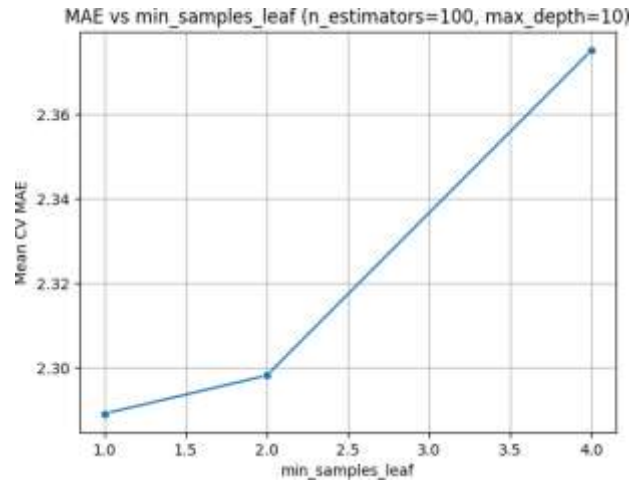


Figure 10. MAE vs. min_samples_leaf

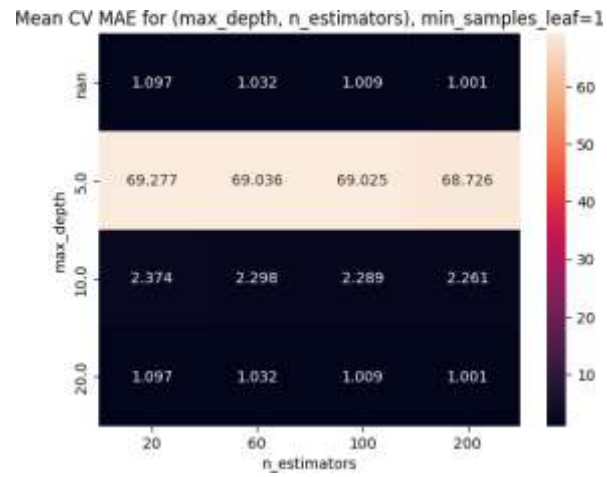


Figure 11. Cross-Validation Grid Results

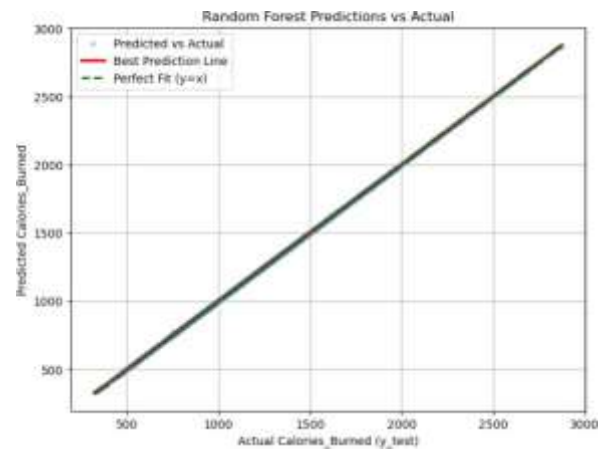


Figure 12. Predictions vs. Actual Calories Burned

4. Interpretation of Key Results

The model achieved test set errors that fall within a reasonable range given the variability in human metabolism and workout intensity. RF successfully captured nonlinear effects. The relatively low MAE indicates the model can predict calories expenditure with consistent accuracy. The predicted vs. actual plot shows a slightly dispersed but generally linear trend, with some widening at higher calories ranges due to natural physiological variability in longer or more intense sessions.

Boosting

AdaBoost

1. Hyperparameter Tuning

AdaBoost was implemented using `AdaBoostRegressor` from `scikit-learn`. Hyperparameter optimization was carried out using `GridSearchCV`:

- `n_estimators`: [50, 100, 150]
- `learning_rate`: [0.01, 0.1, 1.0]

The grid search explored all combinations in this space and selected the configuration that minimized validation error. The best-performing configuration was:

- `n_estimators` = 50
- `learning_rate` = 1.0

This setting corresponds to a moderately sized ensemble of regression trees where each boosting step has a relatively strong corrective effect on previous errors.

2. Cross-Validation Performance

For Adaboost, the cross-validated performance did not indicate a low error.

3. Test Set Evaluation

The tuned AdaBoost model (was then evaluated on the held-out test split. Performance was quantified using MAE and RMSE:

- `MAE` $\approx$ 139.40
- `RMSE` $\approx$ 157.10

This indicates that the tuned AdaBoost model struggled to capture the underlying relationships in the data. The consistency across folds suggests that the model was not severely overfitting, but rather underfitting due to its limited ability to model the complexity of the target variable. Overall,

AdaBoost showed notably weaker performance compared to the tree-based ensemble methods and gradient boosting model.

4. Predicted vs Actual Visualization and Interpretation

A scatter plot of predicted versus actual Calories_Burned was generated using the evaluate_model function as shown in Figure 13.

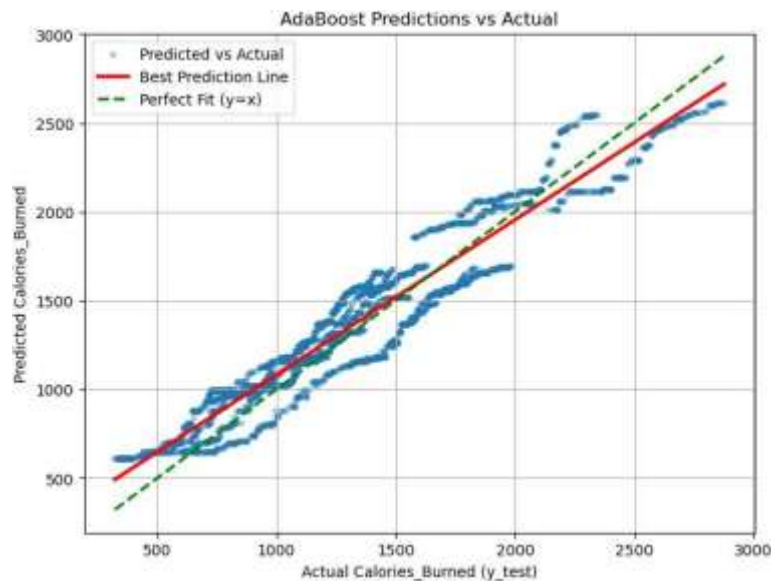


Figure 13. AdaBoost model prediction vs. Actual calories burned

As seen in Figure 13, it includes:

- Blue scatter points representing individual test samples,
- A green dashed line corresponding to the ideal case ($y = x$), and
- A red regression line representing the best linear fit between predictions and ground truth.

The scatter points generally follow an upward linear trend, confirming that AdaBoost captures the overall relationship between input features and caloric output. However, several systematic patterns are visible:

- The red regression line lies below the perfect-fit line for most of the range, indicating that AdaBoost tends to underestimate calories, especially at higher values (e.g., in the 2,000-3,000 kcal range).
- The slope of the best-fit line is less than 1, meaning predicted values increase more slowly than actual values as activity intensity rises.
- The intercept is positive, meaning that the model tends to slightly overpredict for low calorie values and increasingly underpredict as values rise.

- A certain degree of horizontal banding (multiple actual values mapping to similar predictions) is visible, which is characteristic of AdaBoost with relatively shallow base estimators and indicates limited prediction granularity.

Overall, the AdaBoost model learns the core trend reasonably well and achieves acceptable accuracy but exhibits a systematic underprediction bias at the high end of the calorie range. This suggests that deeper base learners or more flexible boosting algorithms may be better suited for capturing higher-order nonlinear interactions in this dataset.

Gradient Boost

1. Baseline Gradient Boosting Model

As an initial experiment, a baseline GradientBoostingRegressor was trained using a learning rate of 0.1 and 60 as the number of trees.

On the test set, this baseline model achieved:

- MAE ≈ 17.98
- RMSE ≈ 23.99

These results already indicate strong predictive performance, with average errors on the order of tens of calories, which is small compared to the overall 300-3000 kcal range in the dataset.

A predicted-versus-actual scatter plot was generated as shown in Figure 14.

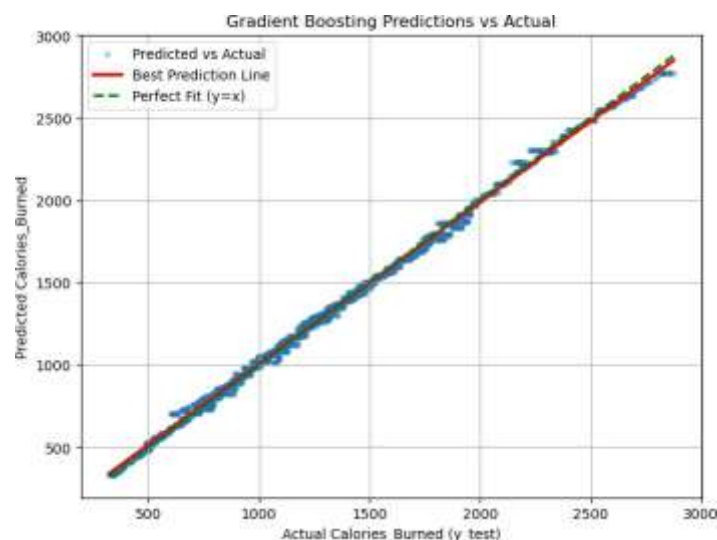


Figure 14. Gradient Boost model prediction vs. Actual calories burned

The figure includes a red best-fit line and the green perfect-fit line ($y = x$). The points clustered closely along the diagonal, suggesting good calibration and relatively low variance in prediction errors even for the untuned model.

2. Hyperparameter Tuning with GridSearchCV

To further improve performance, a more systematic hyperparameter search was performed using GridSearchCV with 3-fold cross-validation and negative MSE as the scoring metric:

- `n_estimators`: [50, 100, 150]
- `learning_rate`: [0.01, 0.1, 1]
- `max_features`: [0.5, 0.7, 1.0]

The search identified the following best-performing configuration:

- `learning_rate` = 0.1
- `max_features` = 1.0
- `n_estimators` = 150

This setting corresponds to a moderately large ensemble with full feature usage at each split and a conservative learning rate, allowing the model to incrementally refine predictions without overfitting.

3. Test Set Evaluation (Tuned Model)

The tuned Gradient Boosting model was evaluated on the same held-out test set. The resulting metrics were:

- `MAE` = 9.7425
- `RMSE` = 13.1186

These values are substantially lower than those of AdaBoost and even the baseline Gradient Boosting configuration. An average absolute error below 10 kcal is extremely small given the wide variation in exercise intensity and duration, indicating that the tuned Gradient Boosting model offers very high accuracy for this prediction task.

4. Predicted vs Actual Visualization and Interpretation

The tuned Gradient Boosting model's performance was further analyzed using a predicted-vs-actual scatter plot shown in Figure 15.

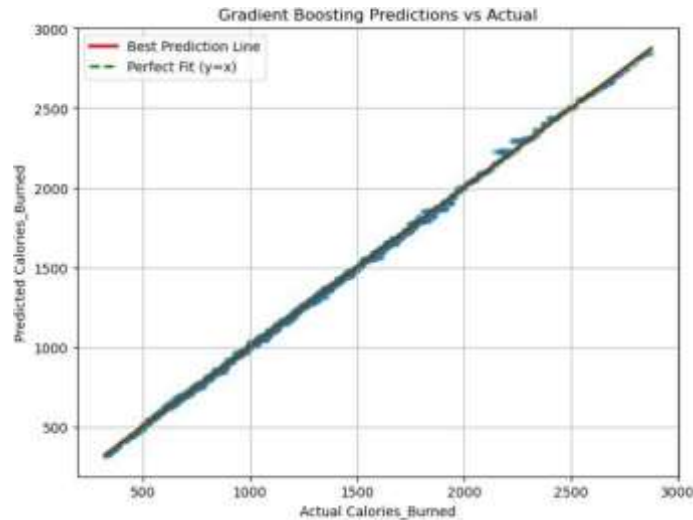


Figure 15. Gradient Boost tuned model prediction vs. Actual calories burned

Key observations are:

- The blue points lie tightly clustered along the diagonal across the full range of Calories_Burned (approximately 350-3000 kcal), indicating low variance and strong calibration.
- Unlike AdaBoost, there is no clear pattern of underprediction at higher calorie levels and no visible banding or compression, suggesting that Gradient Boosting effectively models both low and high calorie regimes.

Overall, Gradient Boosting provided better performance among the ensemble methods tested. The combination of low MAE/RMSE, excellent alignment with the perfect-fit line, and absence of strong bias confirms that Gradient Boosting is relatively well suited for this dataset and task.

DISCUSSION

Support Vector Machine

The Support Vector Machine model provided moderate predictive performance, though its behavior differed from initial expectations. Despite testing a range of nonlinear kernels, the GridSearchCV procedure selected a linear kernel with $C = 1$ and $\epsilon = 0$ as the best configuration. This outcome indicates that, for this dataset, the relationship between features and the target variable is largely linear, and nonlinear kernels such as RBF or polynomial did not offer performance advantages.

Feature scaling remained an essential preprocessing step, ensuring that the SVR model operated consistently across variables with different magnitudes. However, the model's use of integer-encoded categorical variables likely restricted its ability to fully capture categorical patterns, since integer encoding introduces artificial ordering and does not represent category similarity relationships. More expressive encoding techniques such as one-hot or target encoding could potentially improve performance.

With an R^2 of approximately 0.71, along with an MAE around 230 and RMSE near 276, the SVR model was able to capture meaningful trends but still exhibited substantial prediction error. The fitted line in the predicted-vs-actual plot followed the general data trend but showed considerable spread, reinforcing the limited predictive precision.

Overall, the SVM model contributed useful insights and demonstrated acceptable performance, but it did not outperform the ensemble models. The results suggest that exploring richer feature encodings or nonlinear models, or expanding the hyperparameter search space, may further enhance predictive capability.

Random Forest

The Random Forest Regression model proved to be an effective approach for predicting calories expenditure using diverse physiological, exercise, and nutritional features. Several challenges arose during the development of the model: the dataset contained both categorical and continuous variables, many engineered features, and potential non-linear dependencies. Single tree models struggled with this complexity, but Random Forest handled it well due to its ensemble nature and stochastic feature selection.

One challenge involved selecting an appropriate hyperparameter configuration. Too few trees of shallow depth led to underfitting, while excessively deep trees risked overfitting. GridSearchCV helped systematically identify a balance. The results indicate that moderately deep trees ($\text{max_depth} = 20$) combined with a larger ensemble ($\text{n_estimators} = 200$) provided the best performance. Another challenge arose from the engineered features, which exhibit varying scales and distributions. Standardizing the inputs improved model stability and made training more consistent.

The model's performance metrics demonstrate its usefulness for the calorie's prediction problem. MAE and RMSE values indicate that predictions typically fall within a manageable margin of the true values. While some deviation exists at the extremes (very low or high), this is expected due to natural noise in biological and behavioural data. The prediction vs. actual visualization confirms that the model generalizes well, following a near-linear trend along the ideal line.

Overall, the RF approach contributed significantly to solving the calorie prediction problem. It captured complex feature interactions without requiring heavy assumptions about the underlying data distribution. Its robustness and interpretability also make it suitable for practical use, such as

fitness analytics or recommendation systems. With additional tuning or feature refinement, future work could further enhance performance, but the model in its current state provides solid and reliable predictions.

Boosting

AdaBoost

The AdaBoost model demonstrates solid predictive performance, achieving low cross-validated error and reasonable test-set MAE and RMSE. While it effectively captures the overall trend between features and caloric output, it exhibits consistent underprediction across the entire range, has increasing error at higher calorie values, and has prediction banding due to shallow weak learners

These findings indicate that the model has learned the core pattern but may benefit from deeper base estimators or refined tuning to reduce bias and improve performance in higher-value regions.

Gradient Boosting

Overall, the Gradient Boosting model demonstrates higher accuracy compared to AdaBoost. It showed near-perfect calibration, with slope ≈ 1 , and had low error rates and excellent generalization. It showed a decent performance across the full calorie range while there was no systematic bias, unlike AdaBoost. The predictions were smooth and continuous without banding.

Model Comparison

Table 3. Comparing the 4 ML models

Model	MAE	RMSE	Performance Summary
Random Forest (BEST)	0.64	1.17	Almost perfect predictions, best-fit line overlaps with $y=x$, extremely accurate and stable.
Gradient Boosting	9.74	13.12	Strong ensemble model, good calibration, errors higher than RF but still solid.
Support Vector Regression (SVR)	229.97	276.28	Very poor fit for this dataset; large systematic error; significant underprediction and model bias.

AdaBoost	139.4	157.16	Weak performance; shallow learners fail to capture complex relationships; strong underprediction.
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CONCLUSION C FUTURE WORK

In conclusion, this project set out to explore whether machine learning models could accurately predict total calories burned using a large, multi-feature lifestyle dataset. Through a structured methodology that included data cleaning, leakage prevention, feature engineering, and comparative modeling. The project demonstrated that modern supervised learning techniques offer effective tools for predicting total calories burned.

Across all the experiments, the models benefited from the dataset’s diverse mix of variables. Exploratory Data Analysis revealed that behavioural features were stronger predictors of calories burned than physiological features. These insights reinforced the importance of understanding which factors drive metabolic output and provided a strong foundation for model development.

Comparing models demonstrated performance differences between the algorithms. The Support Vector Regression showed stable linear predictive behavior, with the GridSearchCV selecting a linear kernel and achieving moderate accuracy ($R^2 \approx 0.71$). While it captured the main trends in the data, its performance was limited using integer-encoded categorical features. Random Forest offered strong performance, robust generalization, and ability to model nonlinear interactions with minimal parameter tuning. Its ensemble structure effectively balanced variance and bias.

Among the boosting algorithms, AdaBoost produced reasonable accuracy but displayed systematic underprediction due to the limited flexibility of shallow weak learners. The model learned the overall trend but struggled with higher-order interactions, indicating that more complex boosting architectures are better suited for this dataset. Gradient boosting on the other hand seemed to be a strong performer. It achieved a low MAE and RMSE showing near-perfect alignment between predicted and actual values. The model’s ability to iteratively correct errors coupled with optimized hyperparameters allowed it to capture both low-range a high-range calorie patterns with minimal bias.

Overall, the project demonstrates that machine learning can predict calorie expenditure when supported by a good feature set and a structured preprocessing pipeline. The comparative results show that while traditional model like Random Forest offer good baselines, advanced boosting methods, especially Gradient Boosting, provided good accuracy and robustness.

There are several promising paths for future work. One direction is to enhance the representation of categorical variables by introducing one-hot encoding, target encoding, or embedding-based

techniques that avoid imposing artificial ordering. More complex models such as XGBoost, LightGBM or deep learning architectures could also be explored

Another opportunity could be incorporating wearable device signals to provide richer inputs for the prediction. Finally, expanding the evaluation to external datasets would help assess generalizability and support real-world deployment.

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